Weather Forecasting

BUSINESS MACHINE LEARNING I: MIS7620:M50

Under the Guidance of
Prof. Dr. Scott W Vann

Group: 6
Lakshminarayana, Aishwarya
Masini, Vijaya Durga

Introduction



DOES WEATHER PLAY IMPORTANT ROLE IN OUR LIVES?

WHY IS IT IMPORTANT TO KNOW WEATHER CONDITION OF OUR ENVIRONMENT?



YES!!



IF WE WANT TO GO OUT FOR SHOPPING, ATTEND CLASSES, PLAN FOR VACATIONS ETC, WE CHECK THE TEMPERATURE OUTSIDE.

BECAUSE IT REALLY MATTERS! WE WANT TO MAKE SURE WE ARE SAFE OUTSIDE IN A GOOD WARM CONDITION. EVERYONE LOVE TO STAY HEALTHY AND ACTIVE AND ENJOY THE SURROUNDING.

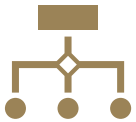
Introduction



So, all of us check the degrees outside, the forecasting is helpful in this job as it collects previous data and predicts future occurrence.



Now we think you all are aware why our team wanted to select this data,



Let’s now understand the data available, where it is from, how we are going to use it and perform the need, and what are lessons learnt and outcomes in simple way.



Business Point of View

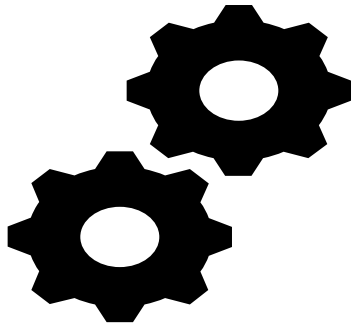
Weather plays prominent role impacting many industries especially manufacturing as predicting future conditions prepare business for risks and optimization of its resource in advance. Weak analysis and forecasting leads to inefficiency and with no proper insights for metrics unpredictability increases, it is important to identify patterns, our project mainly focus to give information for stakeholders to make decision on their business.

Objectives:

- Provide analysis using historical data.
- Give correlations among attributes.
- Help in planning and Decision making.
- Promote cost effective optimization.

Data Source

Link : https://vfl-050.engage.sas.com/SASModelStudio/?projectUrl=/analyticsGateway/projects/6db4775c-4fbb-430f-b947-7b22dc6e1caf&projectTabIndex=1



Software:
Tableau
Python / Collab
SAS VIYA



Data:
Project1Weather Dataset

kaggle 

Source:
Kaggle website



File:
CSV (451KB)

	A	B	C	D	E	F	G	H
1	Date/Time	Temp_C	Dew Point	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather
2	1/1/2012 0:00	-1.8	-3.9	86	4	8	101.24	Fog
3	1/1/2012 1:00	-1.8	-3.7	87	4	8	101.24	Fog
4	1/1/2012 2:00	-1.8	-3.4	89	7	4	101.26	Freezing Drizzle,Fog
5	1/1/2012 3:00	-1.5	-3.2	88	6	4	101.27	Freezing Drizzle,Fog
6	1/1/2012 4:00	-1.5	-3.3	88	7	4.8	101.23	Fog
7	1/1/2012 5:00	-1.4	-3.3	87	9	6.4	101.27	Fog
8	1/1/2012 6:00	-1.5	-3.1	89	7	6.4	101.29	Fog
9	1/1/2012 7:00	-1.4	-3.6	85	7	8	101.26	Fog
10	1/1/2012 8:00	-1.4	-3.6	85	9	8	101.23	Fog

About our Data

Number of Columns: 8

Namely: Date/Time, Temperature in C, Dewpoint Temp in C, Relative Humidity in %, Windspeed in km/hr, Visibility in Km, Pressure in kPa, Weather condition info.

Number of Rows: 8,784

About our Data

The dataset contains date and time at which weather was recorded.

It also have Temperature in degree Celsius helps to know how hot or cold weather was observed.

The Dewpoint temperature gives absolute measure of moisture helps to know how humid it feels.

Relative Humidity percentage shows moisture level in air compared to maximum amount it holds at that temperature.

Windspeed in kilometres per hour shows rate at which air moves horizontally at a point.

Visibility in kilometres helps to know the distance at which object can be seen clearly.

Pressure in kilopascals helps to know how atmospheric pressure influences weather pattern and conditions.

Weather column has textual info description of weather condition.

Tableau Visuals

For our Weather Forecasting data we have made Pie, Area constinuous, Line, Tree map, Gantt, Highlight visuals using Tableau. In short you can see Area continuous graph, Tree map, Highlight table and Pie chart visuals.

Area Continuous graph shows type of weather at that particular wind speed occurring, taking average relative humidity in consideration.

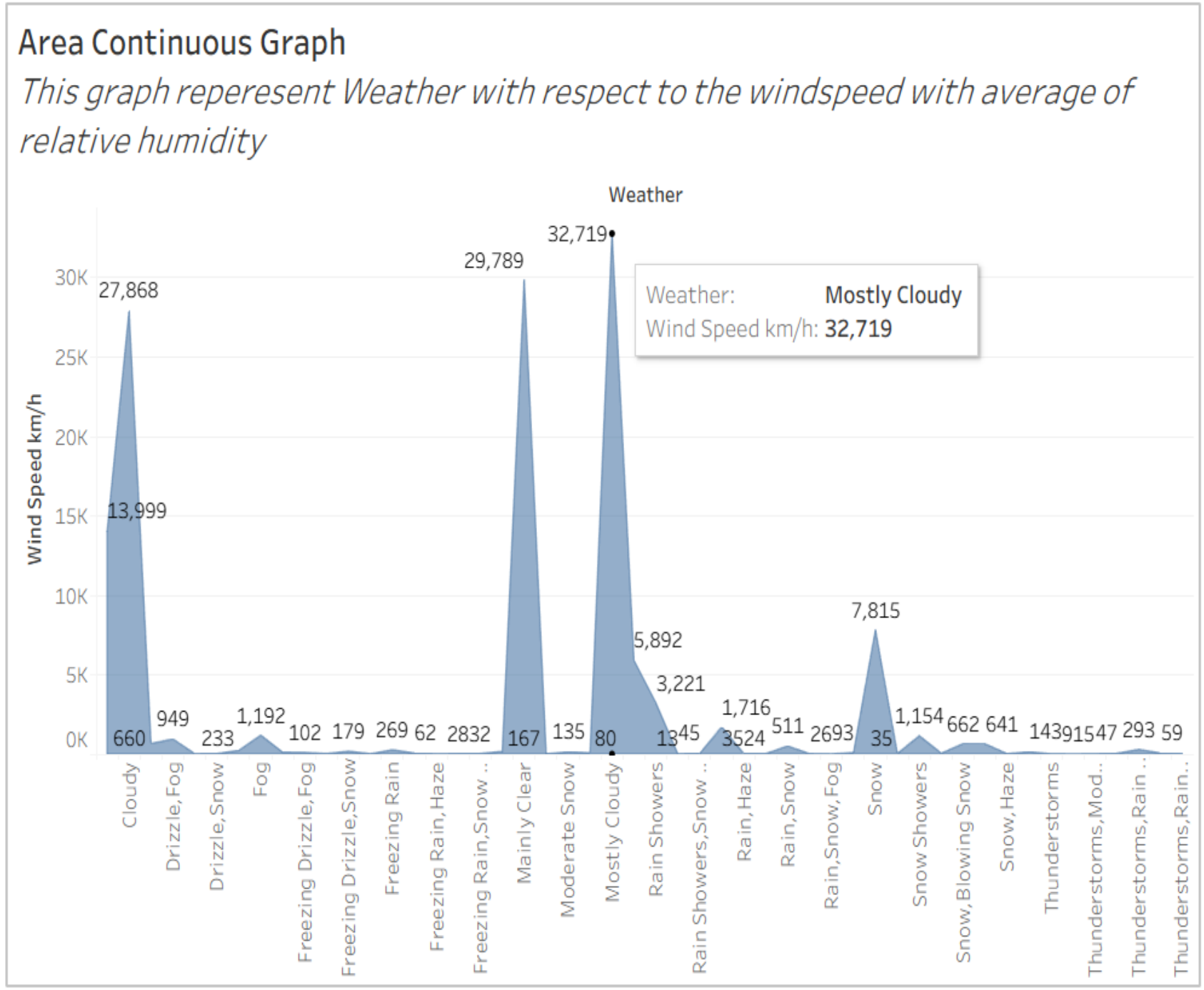


Tableau Visuals

Tree map here Q1 is 1-3 months, Q2 is 4-6 months, Q3 is 7-9 months and Q4 is 10-12 months, and it shows the visibility in terms of average pressure.

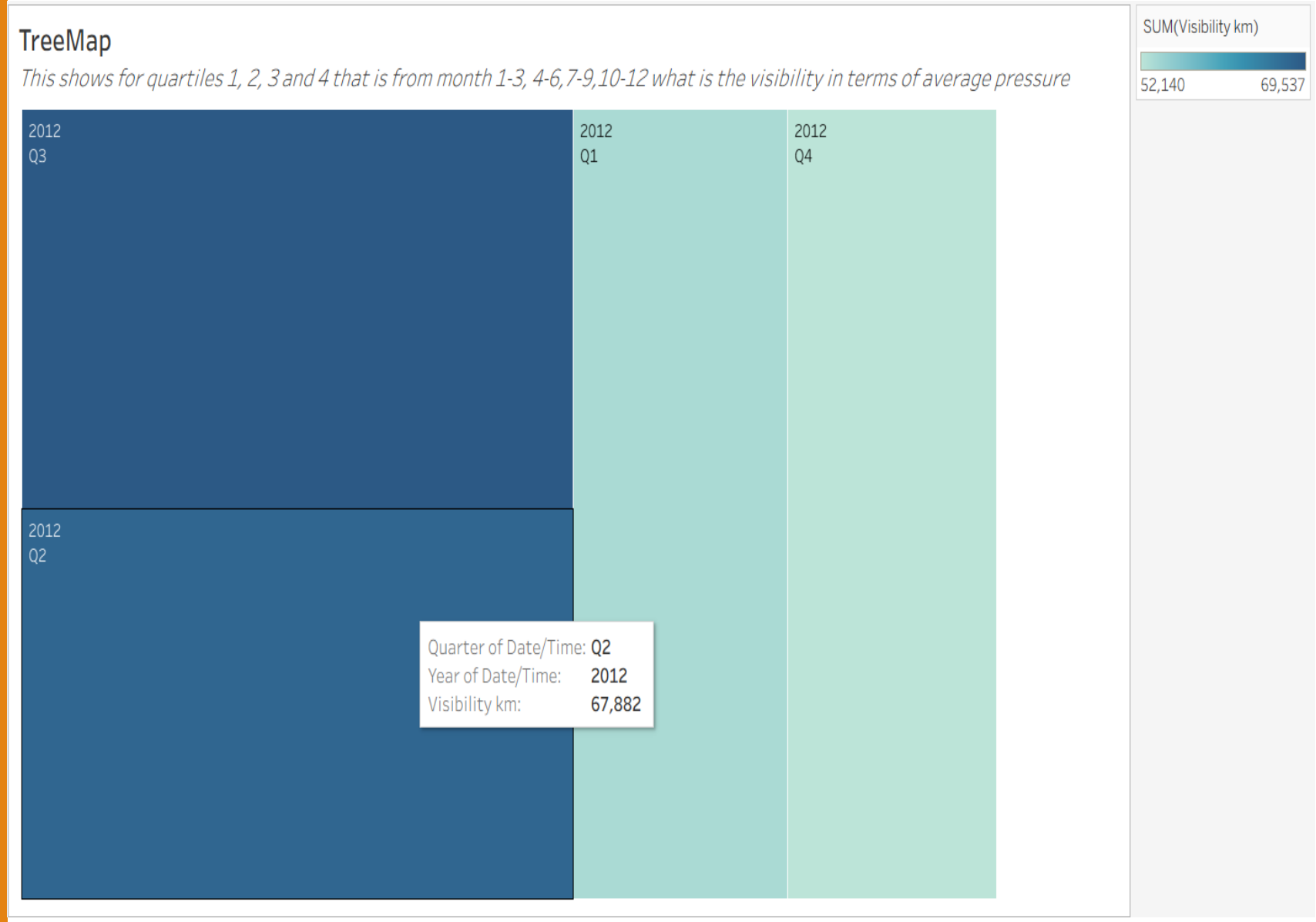
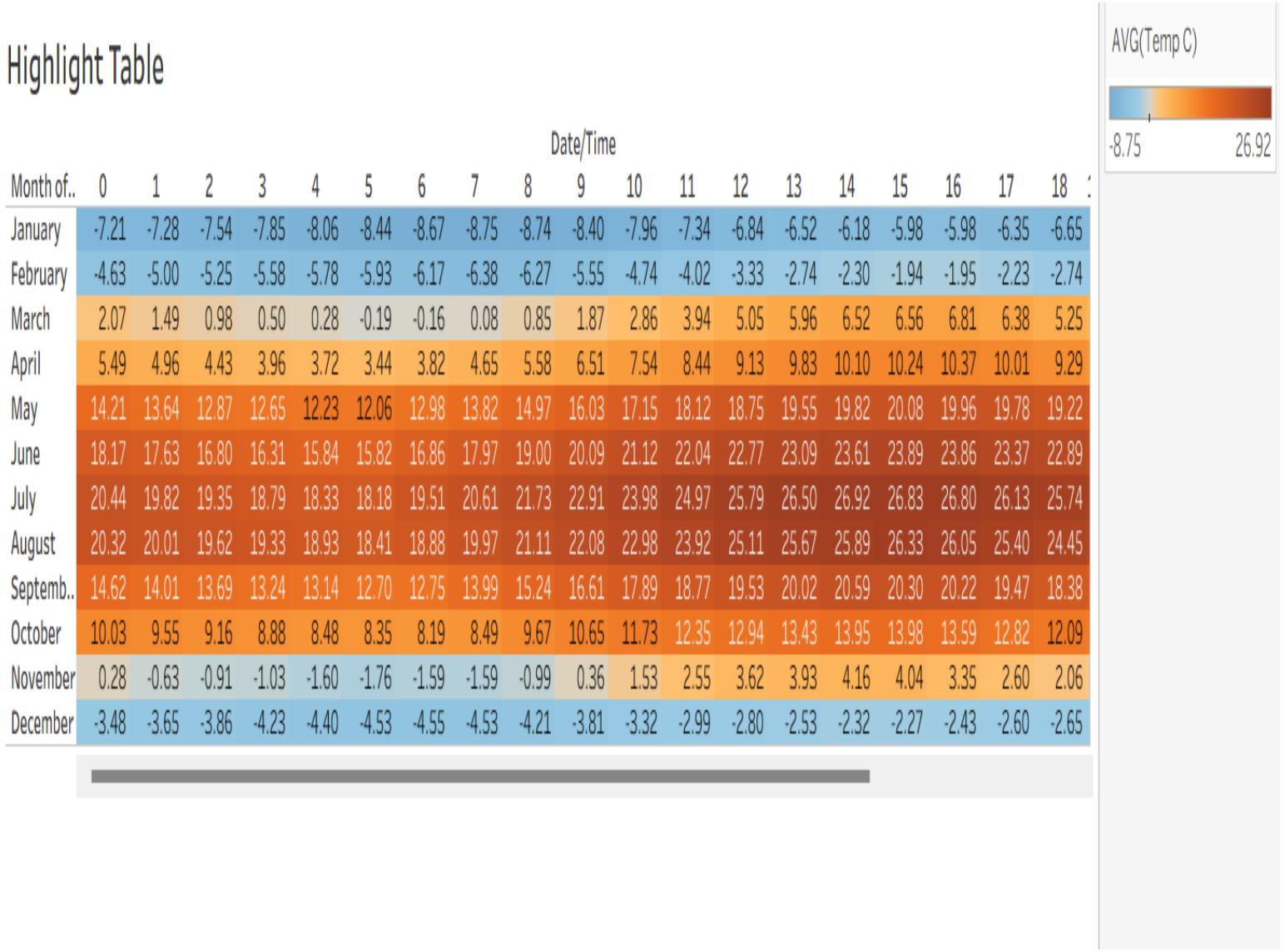


Tableau Visuals

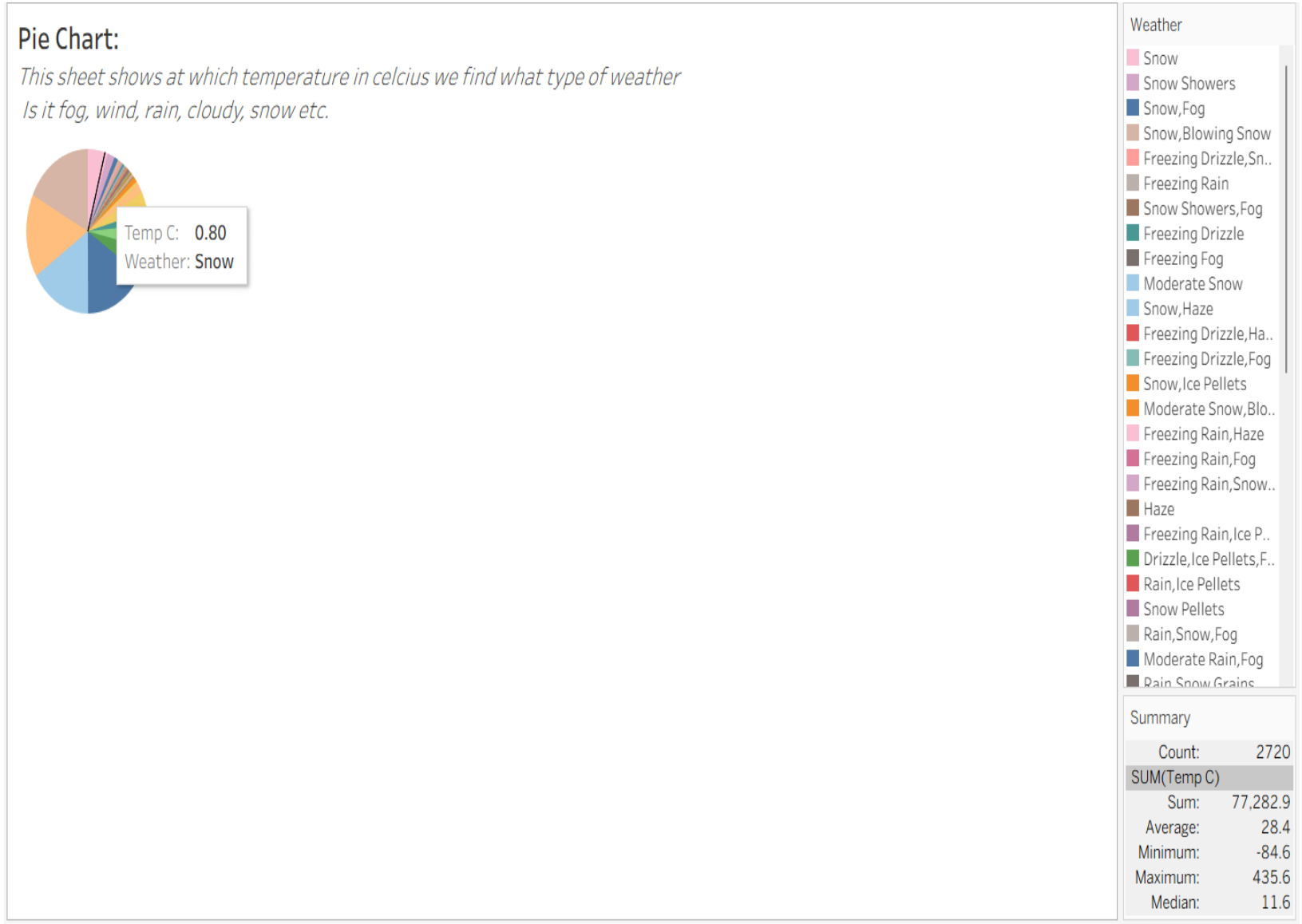
Highlight table here is highlighting the average temperature in Celsius for all months of 24-hour timeline.



Month of..	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
January	-7.21	-7.28	-7.54	-7.85	-8.06	-8.44	-8.67	-8.75	-8.74	-8.40	-7.96	-7.34	-6.84	-6.52	-6.18	-5.98	-5.98	-6.35	-6.65
February	-4.63	-5.00	-5.25	-5.58	-5.78	-5.93	-6.17	-6.38	-6.27	-5.55	-4.74	-4.02	-3.33	-2.74	-2.30	-1.94	-1.95	-2.23	-2.74
March	2.07	1.49	0.98	0.50	0.28	-0.19	-0.16	0.08	0.85	1.87	2.86	3.94	5.05	5.96	6.52	6.56	6.81	6.38	5.25
April	5.49	4.96	4.43	3.96	3.72	3.44	3.82	4.65	5.58	6.51	7.54	8.44	9.13	9.83	10.10	10.24	10.37	10.01	9.29
May	14.21	13.64	12.87	12.65	12.23	12.06	12.98	13.82	14.97	16.03	17.15	18.12	18.75	19.55	19.82	20.08	19.96	19.78	19.22
June	18.17	17.63	16.80	16.31	15.84	15.82	16.86	17.97	19.00	20.09	21.12	22.04	22.77	23.09	23.61	23.89	23.86	23.37	22.89
July	20.44	19.82	19.35	18.79	18.33	18.18	19.51	20.61	21.73	22.91	23.98	24.97	25.79	26.50	26.92	26.83	26.80	26.13	25.74
August	20.32	20.01	19.62	19.33	18.93	18.41	18.88	19.97	21.11	22.08	22.98	23.92	25.11	25.67	25.89	26.33	26.05	25.40	24.45
Septemb..	14.62	14.01	13.69	13.24	13.14	12.70	12.75	13.99	15.24	16.61	17.89	18.77	19.53	20.02	20.59	20.30	20.22	19.47	18.38
October	10.03	9.55	9.16	8.88	8.48	8.35	8.19	8.49	9.67	10.65	11.73	12.35	12.94	13.43	13.95	13.98	13.59	12.82	12.09
November	0.28	-0.63	-0.91	-1.03	-1.60	-1.76	-1.59	-1.59	-0.99	0.36	1.53	2.55	3.62	3.93	4.16	4.04	3.35	2.60	2.06
December	-3.48	-3.65	-3.86	-4.23	-4.40	-4.53	-4.55	-4.53	-4.21	-3.81	-3.32	-2.99	-2.80	-2.53	-2.32	-2.27	-2.43	-2.60	-2.65

Tableau Visuals

Pie chart displays the temperature in Celsius and the weather at that temperature.



Data Exploration and Preprocessing

Before performing any analysis or prediction we have to check if our dataset is clean, we are using Google colab and insert our dataset as bmlweather, we perform following operations:

```
bmlweather.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8784 entries, 0 to 8783
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Date/Time         8784 non-null   object
 1   Temp_C            8784 non-null   float64
 2   Dew Point Temp_C  8784 non-null   float64
 3   Rel Hum_%         8784 non-null   int64
 4   Wind Speed_km/h   8784 non-null   int64
 5   Visibility_km     8784 non-null   float64
 6   Press_kPa         8784 non-null   float64
 7   Weather           8784 non-null   object
dtypes: float64(4), int64(2), object(2)
memory usage: 549.1+ KB
```

```
# looking for satistical summary of numeric variables
bmlweather.describe()
```

	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa
count	8784.000000	8784.000000	8784.000000	8784.000000	8784.000000	8784.000000
mean	8.798144	2.555294	67.431694	14.945469	27.664447	101.051623
std	11.687883	10.883072	16.918881	8.688696	12.622688	0.844005
min	-23.300000	-28.500000	18.000000	0.000000	0.200000	97.520000
25%	0.100000	-5.900000	56.000000	9.000000	24.100000	100.560000
50%	9.300000	3.300000	68.000000	13.000000	25.000000	101.070000
75%	18.800000	11.800000	81.000000	20.000000	25.000000	101.590000
max	33.000000	24.400000	100.000000	83.000000	48.300000	103.650000

We collected information about dataset, then get numeric variable statistic summary.

Data Exploration and Preprocessing

```
# for non-numeric summary
categorical = bmlweather.dtypes[bmlweather.dtypes == "object"].index
print(categorical)
bmlweather[categorical].describe()
```

```
Index(['Date/Time', 'Weather'], dtype='object')
```

	Date/Time	Weather
count	8784	8784
unique	8784	50
top	12/31/2012 7:00	Mainly Clear
freq	1	2106

Get non-numeric variable statistic summary, then frequency for categorical variables.

In our dataset Mainly clear weather category has high frequency of occurrence.

Weather	count
Mainly Clear	2106
Mostly Cloudy	2069
Cloudy	1728
Clear	1326
Snow	390
Rain	306
Rain Showers	188
Fog	150
Rain,Fog	116
Drizzle,Fog	80
Snow Showers	60
Drizzle	41
Snow,Fog	37
Snow,Blowing Snow	19
Rain,Snow	18
Thunderstorms,Rain Showers	16
Haze	16
Drizzle,Snow,Fog	15
Freezing Rain	14
Freezing Drizzle,Snow	11
Freezing Drizzle	7
Freezing Drizzle,Fog	6
Snow,Ice Pellets	6
Snow,Haze	5
Freezing Rain,Fog	4
Moderate Snow	4
Rain,Snow,Ice Pellets	4
Freezing Fog	4
Snow Showers,Fog	4
Thunderstorms,Rain Showers,Fog	3
Thunderstorms,Rain	3
Freezing Drizzle,Haze	3
Rain,Haze	3
Drizzle,Snow	2
Moderate Snow,Blowing Snow	2
Thunderstorms	2
Freezing Rain,Haze	2
Rain Showers,Snow Showers	2
Freezing Rain,Snow Grains	1
Thunderstorms,Heavy Rain Showers	1
Thunderstorms,Moderate Rain Showers,Fog	1
Thunderstorms,Rain,Fog	1
Freezing Rain,Ice Pellets,Fog	1
Moderate Rain,Fog	1
Rain,Snow,Fog	1
Snow Pellets	1
Rain Showers,Fog	1
Drizzle,Ice Pellets,Fog	1
Rain,Ice Pellets	1
Rain,Snow Grains	1

Data Exploration and Preprocessing

```
# lets check for missing values and nonunique values exist or not
bmlweather.isnull().sum()
```

	0
Date/Time	0
Temp_C	0
Dew Point Temp_C	0
Rel Hum_%	0
Wind Speed_km/h	0
Visibility_km	0
Press_kPa	0
Weather	0

```
# nuniqueness
bmlweather.nunique()
```

	0
Date/Time	8784
Temp_C	533
Dew Point Temp_C	489
Rel Hum_%	83
Wind Speed_km/h	34
Visibility_km	24
Press_kPa	518
Weather	50

dtype: int64

```
# lets check for duplicates
bmlweather.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
8779	False
8780	False
8781	False
8782	False
8783	False

8784 rows × 1 columns

dtype: bool

Then we check if our dataset has any missing values, and non-unique values and any duplicates, we found none.

Data Exploration and Preprocessing

```
# lets check for strip whitespace for object columns
bmlweather.duplicated().sum()
```

```
np.int64(0)
```

```
bmlweather['Date/Time'] = pd.to_datetime(bmlweather['Date/Time'])
bmlweather.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8784 entries, 0 to 8783
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype         
---  ------            --------------  -----         
 0   Date/Time         8784 non-null   datetime64[ns]
 1   Temp_C            8784 non-null   float64       
 2   Dew Point Temp_C  8784 non-null   float64       
 3   Rel Hum_%         8784 non-null   float64       
 4   Wind Speed_km/h   8784 non-null   float64       
 5   Visibility_km     8784 non-null   float64       
 6   Press_kPa         8784 non-null   float64       
 7   Weather           8784 non-null   object        
dtypes: datetime64[ns](1), float64(6), object(1)
memory usage: 549.1+ KB
```

```
New 'Weather_Category' column created.
Unique values and their counts in 'Weather_Category' column:
Weather_Category
Cloudy           3797
Clear            3432
Rain              689
Snow              556
Fog               241
Drizzle            51
Haze               16
Thunderstorms       2
Name: count, dtype: int64
```

While checking outliers we found outliers in 4 columns namely Dew point temp_C, Rel Hum_%, Wind Speed_km/h, Visibility_km, Press_kPa, and then we overcome it by applying Capping. We converted Date/Time to datetime for having time-based analysis and feature extraction, also we cleaned Weather column by grouping them into broader categories.

Predictive Modeling

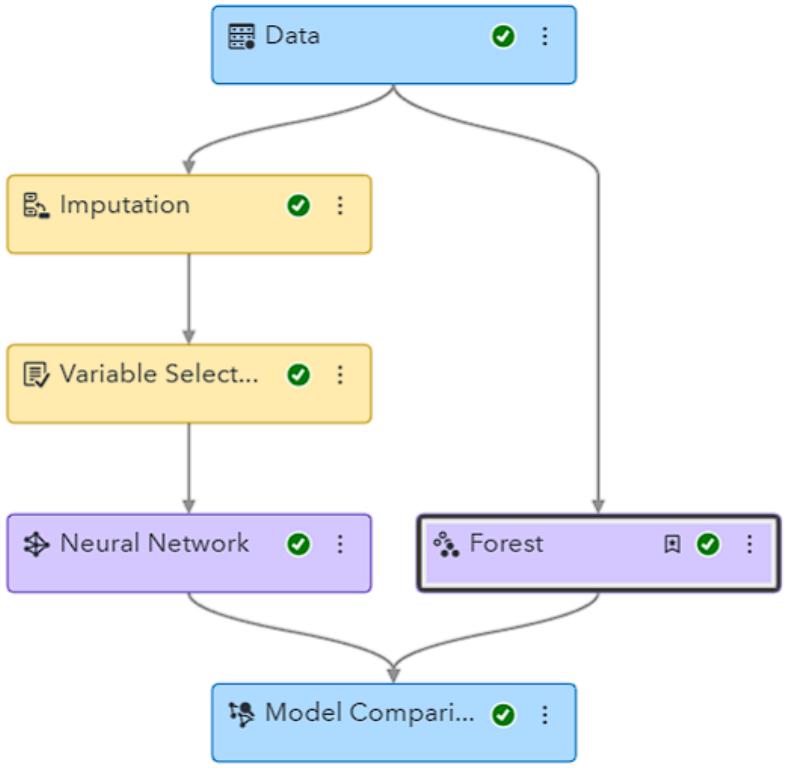
We create 2 models that is Neural network and Forest



We are building End to End Analysis lifecycle, where we use SAS Viya environment and load our dataset which is in csv format, then build model using pipeline, pick champion model and deploy in SAS Model Manager.



Target variable is Temperature_C.



Predictive Modeling

We create pipeline from Data to Model Comparison.

Data: Defines all information about dataset.

Imputation: Imputes missing values for class and interval inputs using specified methods.

Variable Selection: Perform unsupervised and several supervised methods of variable selection to reduce number of inputs.

Neural Network: Fit a fully connected neural network model.

Forest: Fit a forest model consists of multiple decision trees based on different samples of data and subsets of inputs.

Model Comparison: Compares the performance of competing models by using various benchmarking criteria.

Predictive Modeling

Variable Selection result:

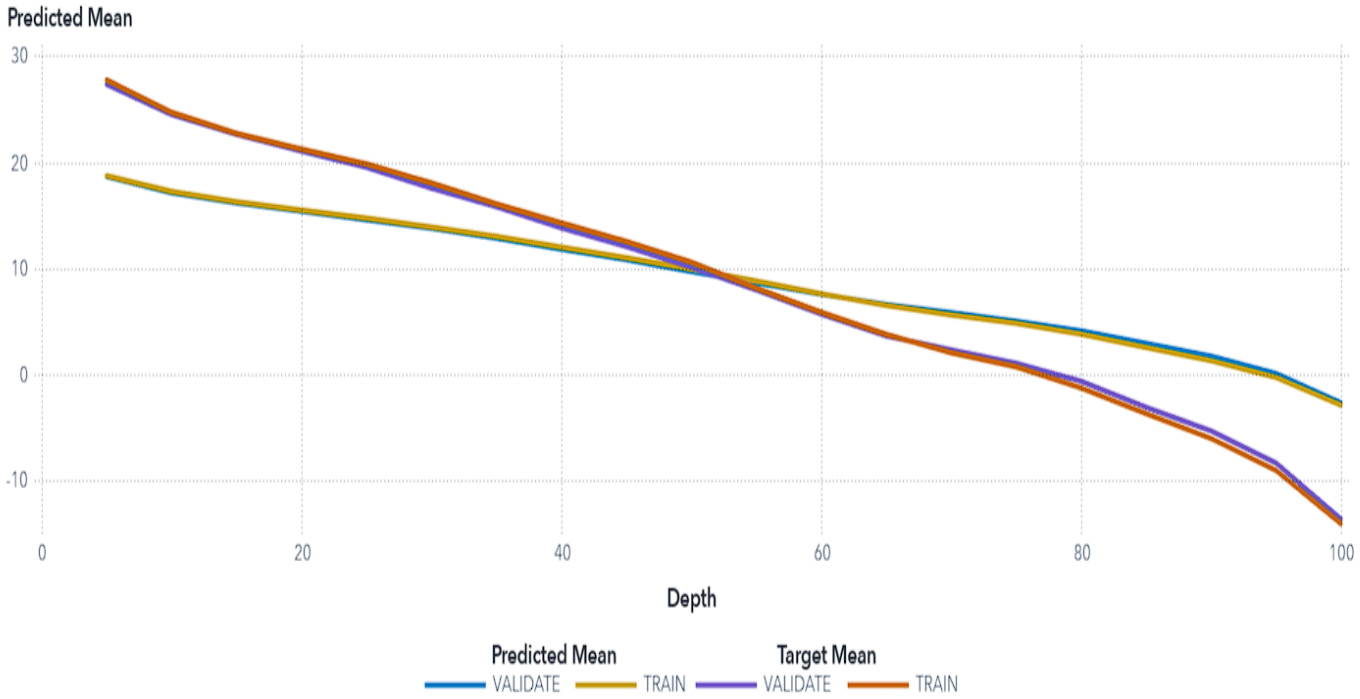
The summary shows, all the three that is Fast, Linear regression and Decision tree rejects the input Visibility_km

Variable Selection Combination Summary

Name	Variable Label	Fast	Linear Regression	Decision Tree	Input	Rejected	Output Role
DEW POINT TEMP_C		INPUT	INPUT	INPUT	3	0	INPUT
PRESS_KPA		INPUT	INPUT	REJECTED	2	1	INPUT
REL HUM_%		INPUT	INPUT	REJECTED	2	1	INPUT
VISIBILITY_KM		REJECTED	REJECTED	REJECTED	0	3	REJECTED
WIND SPEED_KM/H		REJECTED	INPUT	REJECTED	1	2	REJECTED

Predicted Reports

View chart: Actual and predicted by depth



For this plot, each partition of the data is sorted by the predicted target P_Temp_C for the actual target Temp_C, in descending order. The data is then divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the mean of the predicted target and actual target are calculated and plotted for each quantile (depth in increments of 5).

The greatest difference between the actual and predicted target means is 11.177 and occurs for the TRAIN partition at depth 100.

Predictive Modeling

Neural Network results:

Fit Statistics

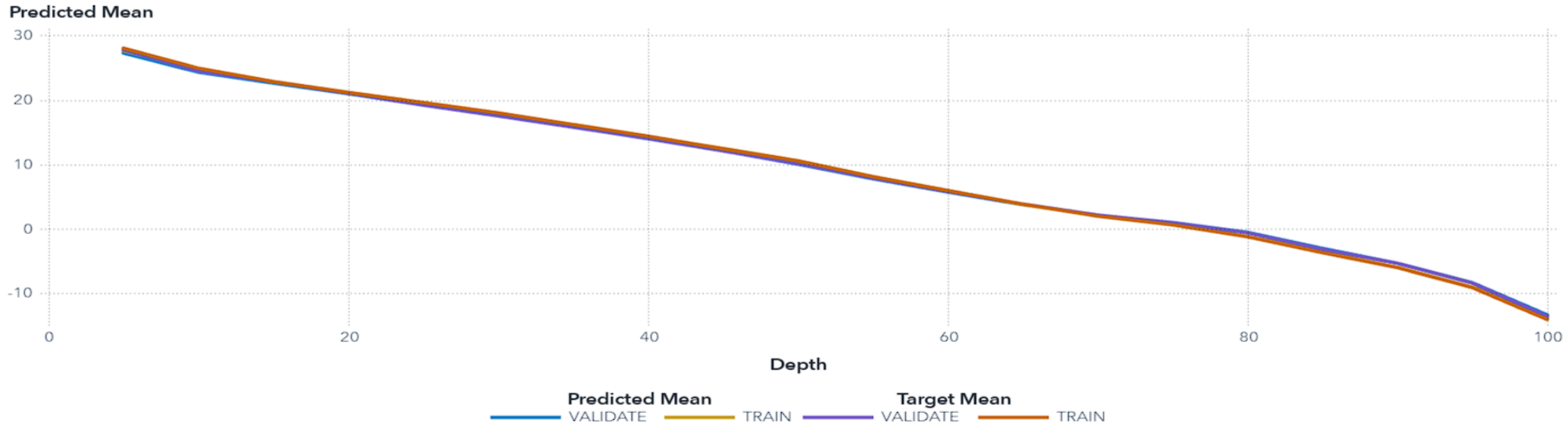
Target N...	Data Role	Partition...	Formatt...	Number...	Average...	Divisor f...	Root Av...	Mean A...	Root Me...	Mean S...	Root Me...	R-Square
Temp_C	VALIDATE	0	0	2,635	31.6339	2,635	5.6244	4.7752	2.1852	0.3895	0.6241	0.7586
Temp_C	TRAIN	1	1	6,149	33.3617	6,149	5.7760	4.9162	2.2173	0.4048	0.6363	0.7599

Predictive Modeling

Forest model results:

Predicted Reports

View chart: Actual and predicted by depth



For this plot, each partition of the data is sorted by the predicted target P_Temp_C for the actual target Temp_C, in descending order. The data is then divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the mean of the predicted target and actual target are calculated and plotted for each quantile (depth in increments of 5).

The greatest difference between the actual and predicted target means is 0.497 and occurs for the VALIDATE partition at depth

Fit Statistics

Target N...	Data Role	Partition...	Formatt...	Number...	Average...	Divisor f...	Root Av...	Mean A...	Root Me...	Mean S...	Root Me...	R-Square
Temp_C	VALIDATE	0	0	2,635	0.4484	2,635	0.6696	0.4091	0.6396	0.0184	0.1355	0.9966
Temp_C	TRAIN	1	1	6,149	0.2748	6,149	0.5242	0.3355	0.5793	0.0152	0.1233	0.9980

Predictive Modeling

Model Comparison results:

Model Comparison

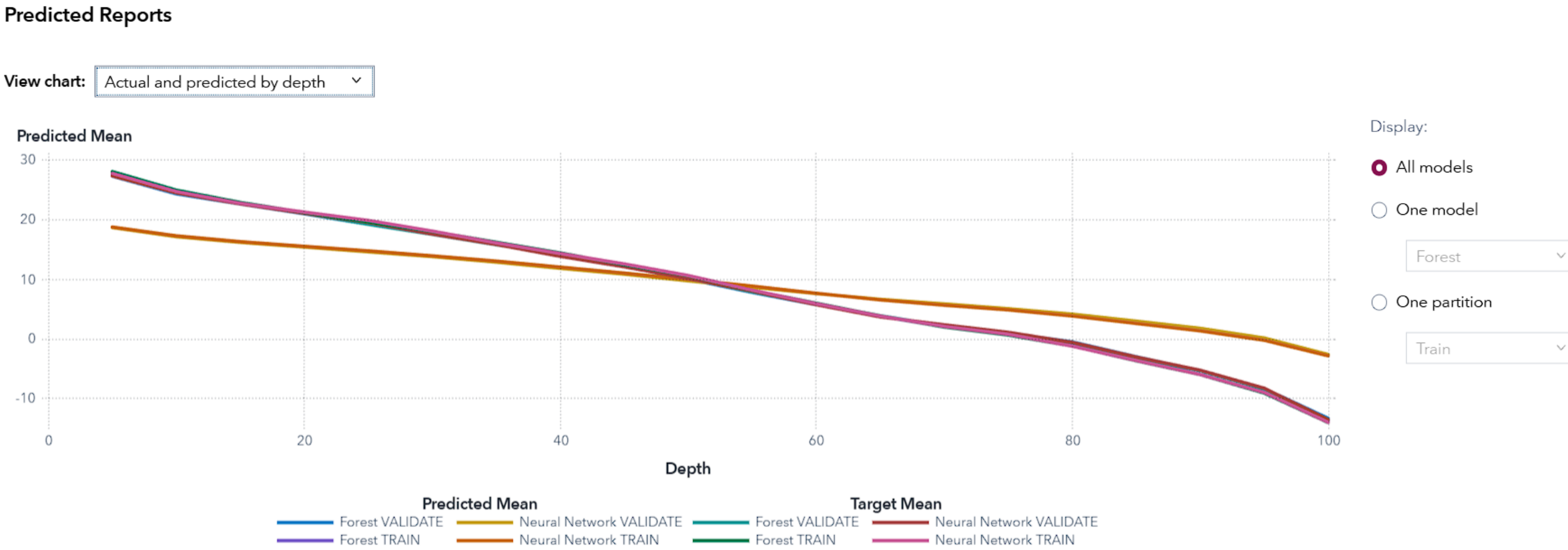
Champion	Name	Algorithm Name	Average Square...	Data Role	Number of Obs...	Root Average S...	Root Mean Abs...	Root Mean Squa...
★	Forest	Forest	0.4484	VALIDATE	2,635	0.6696	0.6396	0.1355
	Neural Network	Neural Network	31.6339	VALIDATE	2,635	5.6244	2.1852	0.6241

Fit Statistics

Statistics Label	Train: Forest	Validate: Forest	Train: Neural Network	Validate: Neural Network
Average Squared Error	0.2748	0.4484	33.3617	31.6339
Divisor for ASE	6,149	2,635	6,149	2,635
Formatted Partition	1	0	1	0
Mean Absolute Error	0.3355	0.4091	4.9162	4.7752
Mean Squared Logarithmic Error	0.0152	0.0184	0.4048	0.3895
Number of Observations	6,149	2,635	6,149	2,635
Partition Indicator	1	0	1	0
R-Square	0.9980	0.9966	0.7599	0.7586
Root Average Squared Error	0.5242	0.6696	5.7760	5.6244
Root Mean Absolute Error	0.5793	0.6396	2.2173	2.1852
Root Mean Squared Logarithmic Error	0.1233	0.1355	0.6363	0.6241
Target Name	Temp_C	Temp_C	Temp_C	Temp_C

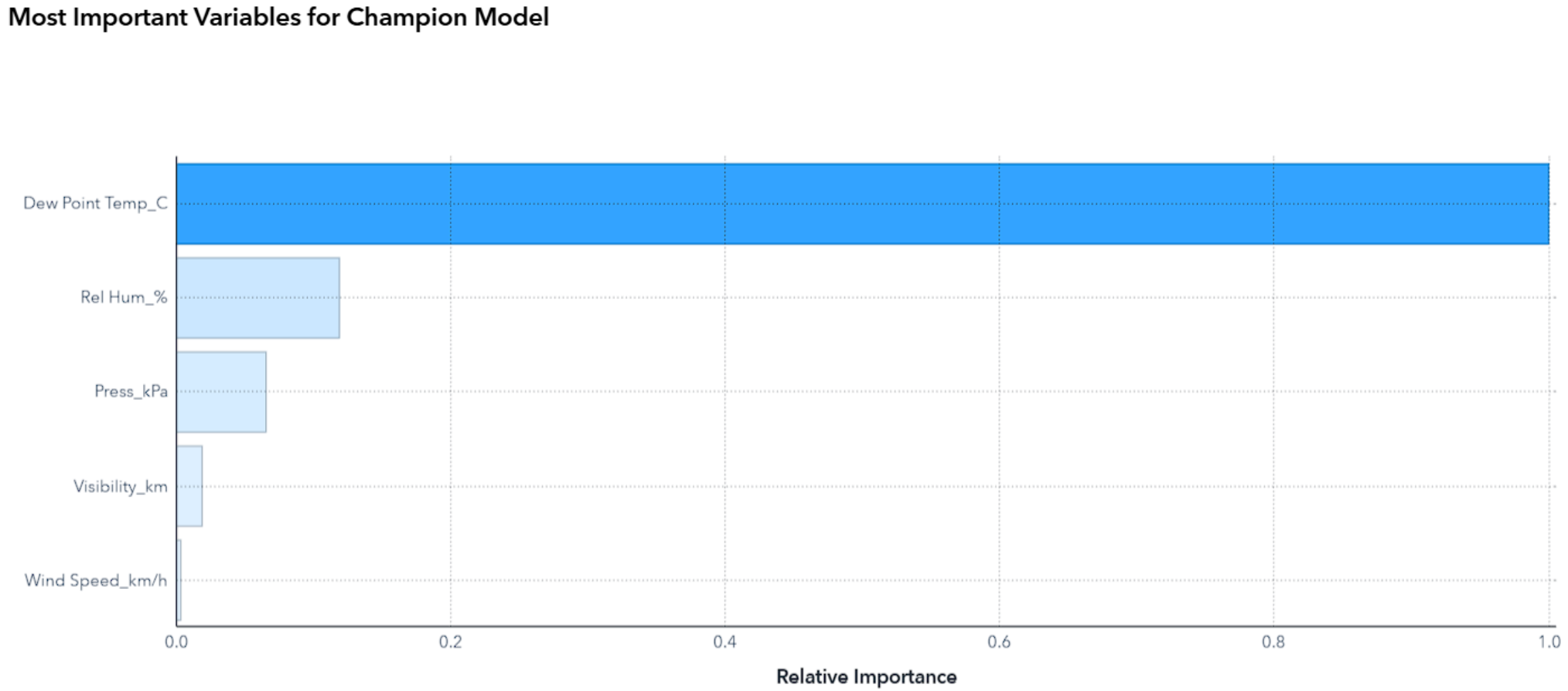
Predictive Modeling

Model Comparison results:



Predictive Modeling

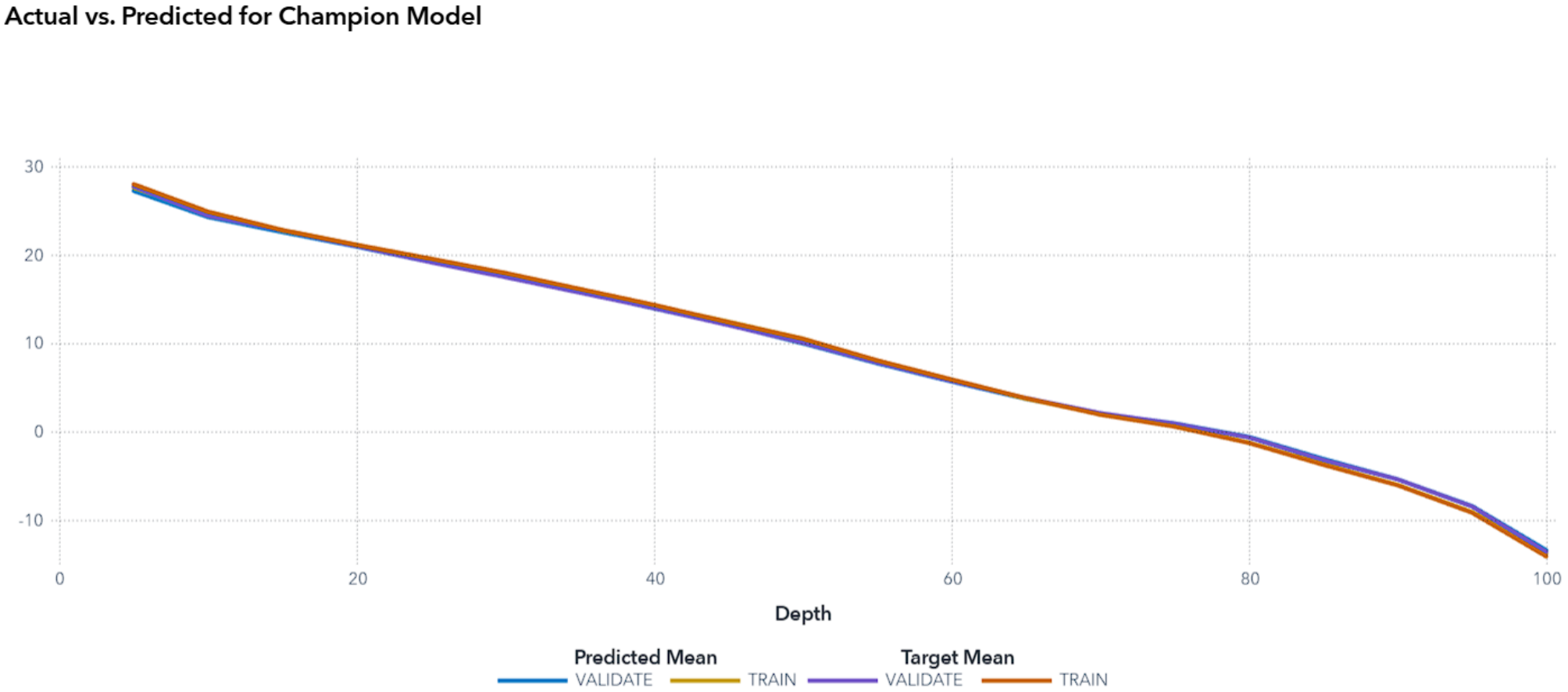
Insights:



This plot shows the 5 most important variables, as determined by the relative importance calculated using the actual model. The most important input for this model is Dew Point Temp_C. The input Rel Hum_% has a relative importance of 0.12, for example, which means it is 0.12 times as important as Dew Point Temp_C.

Predictive Modeling

Insights:



For this plot, each partition of the data is sorted by the predicted target P_Temp_C for the actual target Temp_C, in descending order. The data is then divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the mean of the predicted target and actual target are calculated and plotted for each quantile (depth in increments of 5).

The greatest difference between the actual and predicted target means is 0.497 and occurs for the VALIDATE partition at depth 5.

Predictive Modeling

Interpretation of Finding:

- Forest model is performing best compared to Neural network.
- Dewpoint temperature is appearing consistently meaning it is most important variable.
- Relative Humidity percentage has 0.12 relative importance compared to Dewpoint, as it also influences our prediction quality.
- Other variables have lower predictive contribution.
- The prediction is close to actual values where 0.497 deviation is small than overall temperature range.
- Neural network captured non-linear patterns not effectively as Forest model and also didn't generalize validation partition.

Recommendations:

- Use Forest model for forecasting.
- Improvements can be made from rolling average, weather conditions encoded as categorical levels.
- Weather pattern changes normally due to season changes, so retraining model is highly recommended.

Lessons Learnt & Summary

- Dewpoint temp_C variable gives strongest signal and is dominant predictor.
- Forest model worked extremely well.
- Neural networks require additional features and more training.
- Actual v/s Predicted plots shows prediction is close to actual values across 20 quantiles.
- Model pipeline gives quick comparison between different models (here neural networks and forest)
- No matter which model it is, every model require periodic training and monitoring to have accuracy.

Future Extentions

- The champion model can be deployed for real-time scoring.
- Instead of taking only Temp_C, we can project to multivariate forecasting using different variables.
- Upgrade Neural network with help of new engineering features to make it competitive.
- Extend short time prediction to long range.



Thank you All

Hope you liked our Presentation!